Tools For Data Science

1. **Languages used :**

   TOP 3 :

   Python, R, SQL

2. **INTRODUCTION TO PYTHON**

   For Data Science – computing libraries like Pandas, Numpy, SciPy, Matplotlib

   For AI – PyTorch, TensorFlow, Keras, Scikit learn.

   Natural Language Processing (NLP) : Natural Language Tool Kit

3. **INTRODUCTION TO R LANGUAGE**

   Free Software,

   Similarities between Python and R language

   Both commonly refer to the same set of license

   Both support collaboration

   In many cases these terms can be used interchangeably (but not all)

   Difference –

   Pyhton or open source is business focused while free software or R is value focused.

   - R has become worlds largest repository of statistical knowledge
   - Its has 15000 publicly released package
   - R integrates with other languages like C++ java, .net , python .
   - Mathgematical operation like matrix multiplication work straight out of the box.

- Object oriented programming facilities than other statistical language

4. INTRODUCTION TO SQL(STRUCTURED QUERY LANGUAGE)

20 yrs older than python and R

Used in : handling structured data , i.e. Data incorporating relations among entities and variables

Used in database

Available database – mySQL, ibm db2, oracle database**, SQLite,** MariaDB, PostgreSQL,

5. OTHER LANGUAGE USED IN DATASCIENCE

SCALA, JAVA, C++, JULIA, JS, PHPVISUAL BASIC.

6. DATA SCIENCE TOOLS

a. Open Source Tools for Data Science
   1. Widely used sources (relational database) – **my SQL &** PostgreSQL
   2. Data Integration and Transformation tools- Apache AIRFLOW,kube flow,apache kafka,apache nifi, apache spark SQL , NODE RED.
   3. DATA VISUALIZATION TOOLS – HUE, KIBANA, SUPERSET
   4. MODEL DEPLOYMENT – Prediction IO ,SELDON,MLEAP, TENSOR FLOW SERVICES

5. MODEL MONITORING AND ASSESMENT – ModelDB, PROMETHEUS,
6. DEVELOPMENT ENVIRONMENT – jupyter , jupyter lab, apache zeppelin, R studio,
7. EXECUTION ENVIRONMENT – APACHE SPARK.APCHE FLINK, RAY
8. FULLY INTEGRETED VISUAL TOOLS – KNIME

7. COMMERCIAL TOOLS FOR DATA SCIENCE

a. DATA BASE MANAGEMENT- ORACLE DATABASE, MS SQL SERVER , IBM DB2 P.S. ETL = EXTRACT , TRANSFORM , LOAD
b. DATA INTEGRATION AND TRANSFORMATION – IBM INFOSPHERE DATA STAGE , INFORMATICA , TALEND, WATSON STUDIO DESKTOP
c. DATA VISUALIZATION (BESUINESS INTELLIGENCE TOOLS) – POWER BI ,IBM COGNOS ANALYTICS, TABLEAU.
d. MODDEL BUILDING – SPSS, SAS MINERS
e. MODEL DEPLOYMENT – IBM SPSSCOLLABRATION AND DEPLOYMENT SERVICES
f. MODEL MONITORING AND ASSESMENT
g. DATA ASSET MANAGEMENT – INFORMATICA, IBM INFOSPHARE
h. DEVELOPENT ENVIRONMENT – WATSON STUDIO DESKTOP OF IBM
i. FULLY INTEGRATED VISUAL TOOL – H2O.AI

8. Cloud based tools –

a. FULLY INTEGRATED VISUAL TOOL - WATSON STUDIO DESKTOP, azure machine learning , h2o.ai.
b. DATA BASE MANAGEMENT- amazon DynamoDB, cloudant, CouchDB,
c. DATA INTEGRATION AND TRANSFORMATION- INFORMATICA, IBM DATA REFINERY
d. DATA VISUALIZATION- DATA MEER , IBM COGNO ANALYTICS.
e. MODDEL BUILDING – GOOGLE CLOUD , IBM SPSS, IBM WATSON MACHINE LEARNING , AMAZON SAGEMAKER MODEL MONITER .